A Crypto Observation Layer for Public LLMs: Compiling World Bundles with Quality-Gated Refusal

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Abstract

Claim. Public LLMs fail on crypto markets for want of a reliable *observation layer*, not for want of another trading strategy. We contribute that layer as an anonymized *world-model runtime* (state compiler + quality gate, not a generative simulator): asynchronous multi-band evidence is compiled into a point-in-time (PIT) *world bundle* that public models can consume; when support is thin, a hard refuse flag forces abstention so half-baked context cannot drive decisions. Epistemic observations $O_{j,t} = (x, \tau, q, g, r)$ and compilation Π_t yield $\mathcal{F}_t^{\mathrm{AI}}$; completeness, honesty (B, U, H), and WMI/ACWMI make quality machine-readable; evidence IDs bind actions to disclosed fields. **Evidence.** On identical OOS asset-days, a Compiled bundle makes the market legible: grounding recovers ready/missing/tilt at ≈0.97 vs. ≲0.23 under Raw. The same layer's quality gate is enforceable: across four live public LLMs (temperature 0), Compiled abstains on every thin day (1.0); Ungated (same content, soft text) abstains only ≈0.68; Raw over-trades and loses; Blind refuses with no feed. Full-OOS and band-thick splits reproduce the ranking. This panel never exceeds the production refuse threshold (max WMI ≈ 0.093<0.2), so Compiled's full refusal is a scarce-support stress test of the gate, not a claim the layer never opens. On the open slice (WMI ≥ 0.05; n=1224), live Ungated CE is positive for every vendor (mean ≈+0.29; bootstrap CIs include 0) and a no-LLM rule beats momentum (ΔCE=0.315, p=0.034)—a content lower bound for the compiled fields, not Compiled-open trading. **Falsifiable claim.** Compilation makes disclosed world fields verifiable; hard quality gates enforce thin abstention where soft text does not; open-slice fields are economically nonempty under a no-LLM probe.

CCS Concepts

• **Computing methodologies** → **Artificial intelligence**; *Machine learning*; • **Applied computing** → *Economics*.

Keywords

observation layer, world bundle, quality-gated refusal, public LLMs, cryptocurrency, point-in-time, trustworthy AI, world models, systems

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1 Introduction

Core claim (read this first). Public LLMs fail on crypto *before* strategy: they lack a complete, honest, auditable *market observation layer*. Venue fragmentation, derivatives, on-chain flows, news, and macro vintages arrive on incompatible clocks [13, 15], while generative world models [7, 8, 11] and tool-using finance agents [22–24] typically *assume* a usable observation stream. This paper builds that missing layer—not a new trading agent and not a generative simulator. In one line: *give public LLMs a crypto observation layer that compiles asynchronous evidence into a consumable PIT world bundle, and forces abstention when quality is insufficient so half-baked context cannot harm decisions.*

What we build. We implement the layer as an anonymized market *world-model runtime*: a state compiler from raw multi-band evidence to a PIT-safe *world bundle* for GPT /DeepSeek /GLM /Gemini-class models, with a typed quality gate (`should_ai_abstain`) as the refuse interface. "World model" here means *observation compiler + quality-gated runtime* (Section 3), not a next-state dynamics model. The compiled world must be **complete** (missingness disclosed), **honest** (stale or role-incoherent evidence gated), and **auditable** (actions bound to evidence). Refusal is the safety valve of that layer, not a substitute for seeing the market.

How we evaluate. The layer has two faces on identical OOS days. **RQ2 (seeing):** Does the compiled bundle make analyst answers (ready /missing /tilt) verifiable against PIT ground truth? **RQ1 (gating; primary ranking):** When the world is thin, does a hard quality flag *enforce* abstention across live public LLMs, where soft disclosure does not? Four arms—Compiled (hard flag), Ungated (same content, no flag), Raw (momentum fragment), Blind (no feed). **RQ3 (secondary):** Are the vintaged tilts economically nonempty under a transparent no-LLM rule vs. momentum /buy-and-hold?

Contributions. (1) **Observation-layer semantics:** epistemic observations, Π_t, lag reconstruction, completeness/honesty, WMI/ACWMI quality gates, evidence-bound actions—enough to define the layer, not a pricing theorem. (2) **Runtime:** vintage collectors, Band-PIT previous-close clock, quality-tagged world bundles, shocks O_t, multi-vendor adapter, frozen Compiled /Ungated /Raw /Blind protocol. (3) **Live validation:** Compiled makes ready/missing/tilt answers checkable (≈0.97 grounding vs. Raw 0.04–0.23); the hard gate abstains 1.0 on every thin day while Ungated soft-judges at ≈0.68; Raw over-trades and loses; Blind refuses with no feed; open-slice Ungated CE is a content lower bound, not Compiled-open trading. The estimand is the four-arm ranking of the observation layer, not ΔCE vs. a strongest-agent baseline.

2 Related Work

World models in AI.. World models learn compact states and often dynamics [7, 8, 11]. We address the complementary *observation-side* problem: a PIT-safe, quality-tagged crypto world *before* dynamics or LLM policy. We claim the missing observation layer generative WMs usually assume—not a Dreamer/JEPA simulator.

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LLM agents in finance. ChatGPT return prediction [14], open financial LLMs [21], memory-augmented trading agents [12, 23], and tool-augmented agents [22, 24] improve agent skill and orchestration. Recent ICAIF work targets adjacent layers: FinAgentBench evaluates agentic retrieval for financial QA [1]; FinSearch builds temporal-aware search agents over market and news feeds [18]; FinResearchBench judges long-horizon research agents with logic-tree protocols [19]; FactorMAD uses multi-agent debate for interpretable alpha mining [3]. Those systems improve how agents fetch, debate, and are scored, but still assume a usable observation stream. We study the missing prior step—a PIT-safe *observation layer* that compiles a world bundle and quality-gates action—and evaluate within-model Compiled / Ungated / Raw / Blind under a frozen schema rather than proposing another memory router, retrieval planner, or debate society.

Selective prediction and trustworthy AI.. Abstention can be optimal under noise [4, 5]. We treat refusal as the quality gate of an observation layer, tied to world quality via an explicit runtime field, not classifier confidence alone.

PIT measurement and crypto microstructure. Macro vintages [2] and crypto microstructure [13, 15] motivate multi-band compilation. Bootstrap discipline for the economic probes follows [9, 17, 20]. Asset-pricing ML [6, 10, 16] is adjacent: we validate the economic content of a compiled world for agents, not priced factors.

3 Observation-Layer Semantics

This section defines the observation layer formally (interface semantics, not a pricing theorem): what counts as an observation, how compilation yields a world state, and when quality forces refusal. Readers who want the systems object first may skim to Section 4; the empirical estimand is the four-arm protocol in Section 5.

Definition 3.1 (Epistemic observation). An epistemic observation is

$$O_{j,t} = (x_{j,t},\ \tau_{j,t},\ q_{j,t},\ g_{j,t},\ r_{j,t}) :$$

value, latest-available time, quality, main-view gate, semantic role. A bare feature dump that omits (τ, q, g, r) is not a world observation.

Definition 3.2 (Financial world state). $W_t = \Pi_t(\mathcal{F}_t^{\text{raw}})$ with scalar quality $q(W_t)$ (e.g. WMI/ACWMI) gating abstention; $\mathcal{F}_t^{\text{AI}} = \sigma(W_t)$.

Definition 3.3 (Compilation). $\Pi_t = B_t \circ M_t \circ A_t$: align clocks, apply honesty/missingness gates, assemble band views.

Completeness, honesty, quality. Ready/limited/missing counts disclose coverage. $\text{WMI}_t = B_t U_t H_t$; ACWMI supports regime-conditional gating. The Compiled layer exposes the quality-gate flag (`should_ai_abstain`) and `thin_world` as first-class fields of the world bundle.

Assumption 3.4 (Bounded lag and increments). Band j has max observation lag Δ_j; latent state increments satisfy $\|S_u - S_{u-1}\| \le \bar{\delta}$.

Proposition 3.5 (Lag reconstruction bound). *Under Assumption 1,*

$$\|\widetilde{S}_t - S_t\| \ \le\ C_{\text{del}}\bar{\delta} + C_{\text{noi}}\varepsilon_t + C_{\text{miss}}m_t.$$

Compilation cannot erase delay; it can refuse to pretend the bound is zero.

Intuition. Crypto agents that concatenate tool dumps silently absorb delay and missingness into the feature vector. Prop. 3.5 makes those terms explicit: a thicker raw payload can worsen the reconstruction if honesty collapses.

Proposition 3.6 (Compilation ≠ feature dump). *Enlarging raw span without Π_t need not enlarge usable $\mathcal{F}^{\text{AI}}$: ungated stale evidence can raise volume while lowering honesty/stability.*

Proposition 3.7 (World-conditional abstention). *If non-abstain actions exceed a world-dependent cost $c_{\text{abs}}(W_t)$, the Bayes action is abstain; with costs decreasing in WMI, abstention forms a lower set in world quality [4].*

Proposition 3.8 (LOBO content vs. gating). *Deleting band j changes value through a content channel and a gating channel. Empirically we report content share under an ungated rule so the probe speaks to nonempty fields.*

Implication for the runtime. Props. 3.6–3.7 say the observation layer must export gates and abstain flags, not tilts alone: compilation without quality disclosure is still a feature dump; disclosure without a hard gate still lets thin worlds drive action. Prop. 3.8 motivates the economic check: if deleting a durable band does not change actions, the bundle is empty of content; the probe shows that is not the case here.

Remark 3.9 (Not a generative WM). *We do not learn $p(s_{t+1} \mid s_t, a_t)$. We deliver an* observation layer*: state compiler + quality-gated refuse runtime.*

4 Runtime Architecture

We describe the anonymized prototype as a layered system (Figure 1; module map in Table 1). Source paths and product names are omitted for double-blind review; an anonymized artifact will be released upon acceptance.

4.1 Layered design

(1) **Data layer (collectors).** Exchange, macro, and alternative collectors write vintage-aware history: observation timestamps plus macro `available_at` vintages.
(2) **Store layer.** Embedded analytical stores hold raw series, merged market panels, and pipeline outputs (readiness, WMI / ACWMI, paper world snapshots).
(3) **Logic layer (compilation).** BandPIT reconstructs band readiness on a previous-close clock; honesty and missingness gates plus band assembly implement Π_t; world-quality indices and availability shocks O_t are first-class.
(4) **Service layer.** A read API serves quality-tagged AI bundles and health/readiness objects to consumers without requiring collector restarts for newly arrived rows.
(5) **Consumer layer.** Frozen-prompt LLM / rule agents call an OpenAI-compatible adapter (temperature 0) under Compiled, Ungated, or Raw treatments.

4.2 Data flow: inputs, processing, outputs

Table 2 traces the runtime end-to-end. *Inputs*: the evaluation archive populates three public bands—exchange klines, funding, and open

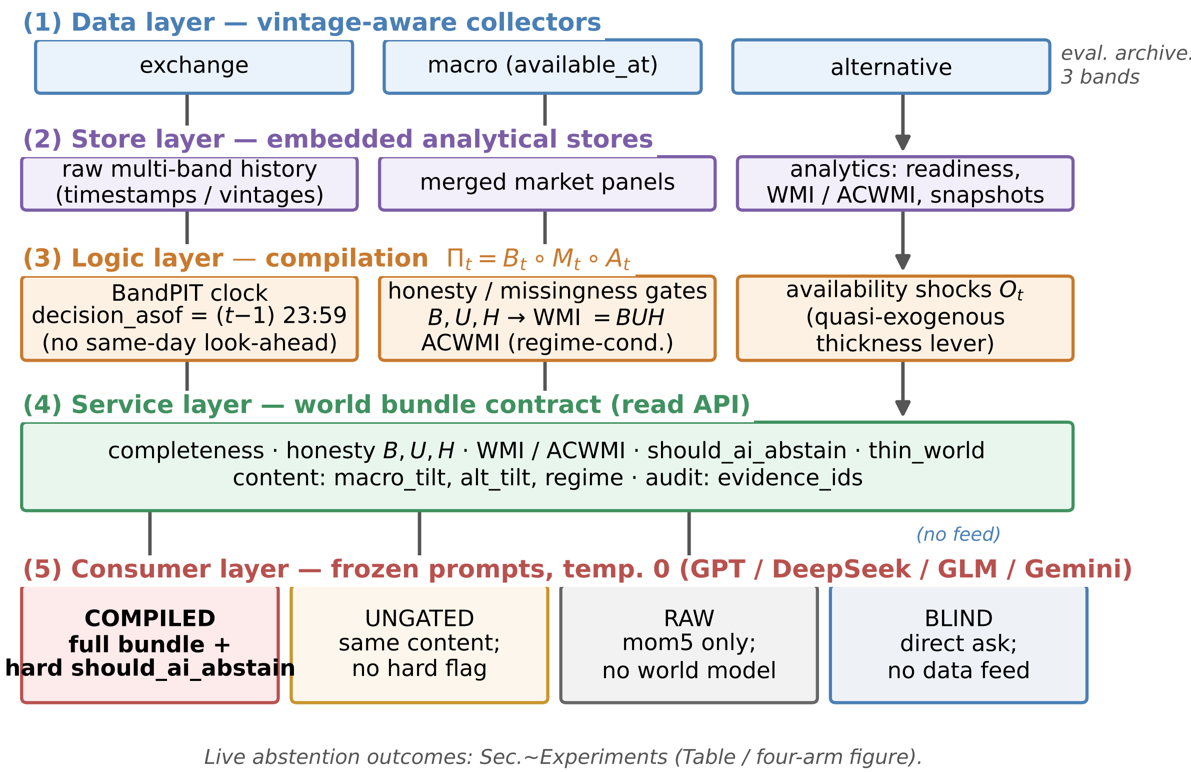

Figure 1: Anonymized observation-layer runtime. Five numbered stages map raw multi-band evidence to a consumable world bundle; the consumer stage defines the frozen four-arm protocol (Compiled / Ungated / Raw / Blind). Live seeing and gating outcomes are in Sec. 5; Blind has no feed from the service stage.

Table 1: Anonymized module map (systems view).

Module	Responsibility
Collectors	Ingest multi-band series with timestamps / vintages
Vintage stores	Persist raw, market, and analytics objects
BandPIT compiler	Previous-close readiness, Π_t, WMI/ACWMI
Bundle builder	Assemble complete/honest/auditable JSON world state
Shock index O_t	Query availability outages for quasi-exogenous levers
LLM adapter	OpenAI-compatible chat; frozen prompts; temp. 0
Eval harness	Compiled/Ungated/Raw/Blind sampling, transcripts, tables

interest via REST; daily macro risk indices (equity volatility, dollar index) with `available_at` vintages; and aggregate stablecoin circulation—each row stored with its observation timestamp. *Processing*: for every asset-day, BandPIT selects the latest observation with timestamp $\leq (t-1)$ 23:59, grades it fresh/stale/missing against per-band age thresholds, aggregates breadth/stability/honesty into WMI and regime-conditional ACWMI, derives content tilts (5-day macro-index changes; 7-day stablecoin-flow slope; return momentum), and logs availability shocks O_t. *Outputs*: a ~3,990-row PIT panel (10 assets × 399 days), per-day world bundles scoped to these three archive bands (Listing 1), and the consumer transcripts/tables behind Section 5.

4.3 PIT previous-close clock

For calendar day t, the decision information set is reconstructed at $(t-1)$ 23:59. The payoff is the same-calendar-day close-to-close return r_t. This removes same-day look-ahead in band statuses and vintaged macro/alternative content. The bundle records the clock in `decision_asof`, and every live run pins `timing_protocol` to the previous-close protocol.

4.4 World bundle (consumer object)

Table 3 lists field groups. Listing 1 shows a compact Compiled example from the OOS panel: exchange missing so 2/3 archive bands ready, WMI low, `should_ai_abstain=true`, and `evidence_ids` binding actions to disclosed bands/tilts. The bundle is what the LLM *sees*; the hard boolean is how the layer *gates* action when that view is thin. Ungated removes the hard boolean (and threshold) but keeps numeric WMI/ACWMI and completeness. Raw retains only `mom5` (plus a non-informative noise bit).

Table 2: Data flow through the runtime (three-band evaluation archive).

Band	Fetched (raw)	Emitted (bundle)
exchange	OHLCV klines (1h/1d), funding, open interest, orderbook	mom5; band status/age; funding context
macro	equity-vol and dollar indices; vintaged macro series	macro_tilt (5d changes); status/age
alternative	aggregate stablecoin circulation	alt_tilt (7d flow slope); status/age
payoff	daily closes (external reference)	close-to-close r_t for scoring only
derived	—	B, U, H, WMI, ACWMI, `should_ai_abstain`, `thin_world`, O_t, `evidence_ids`

Table 3: World-bundle field groups (observation layer output).

Group	Fields
Timing	`decision_asof`, previous-close protocol
Completeness	$n_{\text{ready/limited/missing}}$, band statuses
Honesty	B, U, H, main-view / stale gates
Quality	WMI, ACWMI, `should_ai_abstain`, `thin_world`
Content	macro_tilt, alt_tilt, regime, cascade
Audit	`evidence_ids`, EAR required

Listing 1: Compact Compiled bundle example (OOS exchange gap).

```
{"date":"2026-05-11","asset":"ADA",
 "decision_asof":"2026-05-10T23:59:00",
 "completeness":{"n_ready":2,"n_missing":1},
 "world_model_index":{"wmi":0.015,
   "should_ai_abstain":true,"thin_world":true},
 "macro_tilt":1.0,"alt_tilt":-1.0,
 "audit":{"evidence_ids":[
   "band:exchange:missing","band:macro:ready"]}}
```

4.5 Four information arms and LLM adapter

- **Compiled (full observation layer):** full world bundle; frozen prompt *requires* abstain when `should_ai_abstain` is true.
- **Ungated (disclosure without the gate):** same content, completeness, and numeric WMI; *no* hard flag; soft prompt asks the model to judge thinness (seeing without enforcement).
- **Raw (thin integration):** `mom5` only; no quality index or abstention guidance. This is a common price-only wiring, not an exhaustive tool-agent baseline: the primary estimand is the four-arm ranking of the observation layer (hard gate vs. soft vs. fragment vs. none), not ΔCE against the strongest possible agent.
- **Blind (direct ask):** "Today is d. How should I trade a?" with *no* data feed.

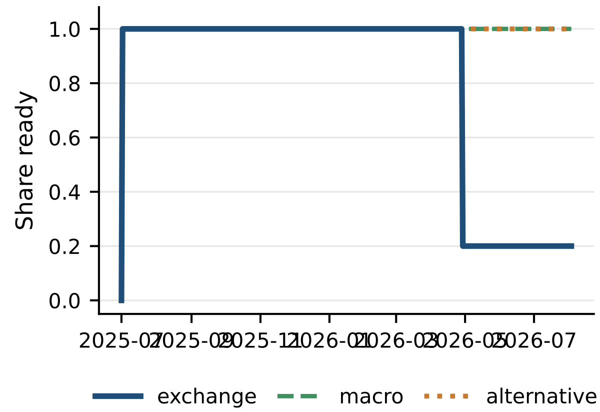

Figure 2: PIT readiness for the three-band evaluation archive (solid / dashed / dotted for grayscale).

Actions $\in$ {bullish, bearish, neutral, abstain}. Primary metrics are thin-world abstain rate and EAR proxy; CE is reported as avoided loss under refusal, not as an alpha claim. The evaluation archive is the three-band set {exchange, macro, alternative} (Figure 2): macro and alternative stay ready throughout; exchange readiness drops in the later OOS window and defines the within-archive *band-thick* contrast below. The refuse rule sets `should_ai_abstain` when WMI < 0.2. On this scarce OOS panel max WMI ≈ 0.093, so every day is thin and Compiled never opens—a scarce-support stress test of the quality gate, not a claim the observation layer never allows action. The adapter uses OpenAI-compatible chat completions; credentials stay in environment variables (never in artifacts).

4.6 Frozen prompt protocols

Listings 2–3 excerpt the frozen Compiled and Ungated prompts (temperature 0; action schema identical). Both consume the same world-bundle content; the only intentional difference is hard vs. soft quality gating. Raw prompts omit world-quality language entirely.

Listing 2: Compiled observation layer (excerpt).

```
Decision rules (excerpt):
1. If world_model_index.should_ai_abstain is true
   OR the world is thin, you MUST "abstain".
2. Prefer band content over raw momentum
   (macro_tilt, alt_tilt are PIT-safe).
3. Rationale must be consistent with evidence_ids.
```

Listing 3: Ungated soft guidance (excerpt).

```
Decision guidance (soft):
1. Judge whether missingness / low WMI/ACWMI
   makes action unsafe; you MAY "abstain".
2. Prefer band content when you act.
NO hard abstain boolean; NO forced policy.
```

Why the hard quality gate is not a confound. Compiled abstention is partly prompt obedience, and that is intentional: the observation layer's job on thin days is to *block* action, not to hope the model discovers thinness (Prop. 3.7). Ungated measures soft judgment from the same disclosed bundle; Raw measures the common price-signal-only wiring. The gating result is that the hard flag is reliable

Table 4: Live abstention rates on identical 100 OOS asset-days (Wilson 95% CIs); Blind abstains 1.0 for every model. ΔCE is avoided loss, not an alpha claim.

Model	Thin C	Thin U [CI]	Abs. R [CI]	ΔCE
gpt-5.4-mini	1.00	0.63 [.53,.72]	0.04 [.02,.10]	+0.73
deepseek-v4-flash	1.00	0.81 [.72,.87]	0.75 [.66,.82]	+1.22
glm-5.2	1.00	0.86 [.78,.91]	0.11 [.06,.19]	+1.04
gemini-3.5-flash-lite	1.00	0.43 [.34,.53]	0.07 [.03,.14]	+1.30
Mean	1.00	0.68	—	+1.07

across vendors; soft refusal is not—which is why half-baked context remains dangerous without the gate.

4.7 Service surface and reproducibility

The service layer exposes read-only objects for the consumer harness: quality-tagged bundles, readiness / WMI–ACWMI snapshots, shock queries O_t, and health metadata. For review we freeze prompts, temperature, action schema, IS/OOS cut, and sampling seed; model IDs and gateway URL are configuration, not estimands. Credentials never enter artifacts. Frozen prompts, protocol notes, and table builders for every reported figure/table ship with the submission; an anonymized harness is available to reviewers on request and public on acceptance.

5 Experiments

5.1 Setup

PIT panel ∼399 days, 10 liquid names, previous-close clock, chronological IS/OOS 200/200 days (≈2,000 OOS asset-days). Live RQ1 uses the *same* stratified 100 OOS asset-days across arms (seed fixed), temperature 0, frozen prompts, and an OpenAI-compatible gateway; a checkpointed full-OOS sweep tests population scale. Vendor models are GPT-, DeepSeek-, GLM-, and Gemini-class flash/mini IDs listed in Table 4. Scarce/thin labels bind on OOS by design; Sec. 5.4 covers fully populated days.

5.2 RQ1 (primary ranking): Quality gate of the observation layer

Table 4 and Figure 3 report the gating face of the layer; all arms share the same 100 OOS asset-days. Under Compiled, all four models abstain on every thin-world day (thin-abs. 1.0); rationales cite world-bundle fields on 93–99% of days. This is quality-gate obedience by design, not spontaneous thinness discovery. Under Ungated, the *same* disclosed bundle without the hard flag yields mean thin-abs. 0.68 (range 0.43–0.86): DeepSeek and GLM stay above 0.80, GPT drops to 0.63, and `gemini-3.5-flash-lite` falls to 0.43 and trades. Unconditional Ungated CE (abstain days as cash) spans −1.04 to +0.02; restricting to days the model actually acts, three of four vendors post large negative act-conditional CE (mean ≈−1.9), so soft judgment does not safely convert a half-gated observation into profitable action. Under Raw, abstain rates are 0.04–0.75 with substantial directional mass (Table 5). Mean ΔCE (Compiled−Raw) is +1.07 and positive for every vendor, because the gated layer sits out losses that Raw incurs on sparse support.

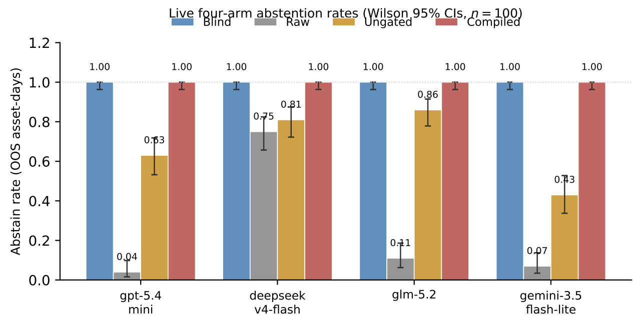

Figure 3: Live four-arm abstention rates (same 100 days; Wilson 95% CIs). Blind refuses with no feed; Raw over-trades on a fragment; Ungated soft-judges a disclosed bundle; Compiled enforces the observation layer's quality gate.

Table 5: Raw-arm action mix (100 days). Compiled abstains on all thin days; Blind (direct ask, no feed) abstains 100% for every model.

Model	Abs.	Bull.	Bear.	Neut.
gpt-5.4-mini	0.04	0.28	0.38	0.30
deepseek-v4-flash	0.75	0.14	0.04	0.07
glm-5.2	0.11	0.40	0.48	0.01
gemini-3.5-flash-lite	0.07	0.44	0.49	0.00

Full-OOS replication. To check that the 100-day ranking is not a subsample artifact, we checkpointed the three data arms on the *full* OOS panel (n=2000 asset-days; every OOS day is thin-world under the scarce-support design). Compiled thin-abstain is 1.000 [.998, 1.000] for all four vendors. Ungated rates tighten the Wilson intervals, with mean thin-abs. 0.56 (GPT 0.666 [.645, .686]; GLM 0.679 [.654, .703] on n=1402 completed calls; DeepSeek 0.540 [.518, .561]; Gemini 0.359 [.338, .380]). Vendor ranking within Ungated shifts relative to the 100-day sample (DeepSeek is less cautious at scale), but no vendor matches Compiled. Where Raw is trusted, over-trading remains sharp: GPT and Gemini abstain at 0.000 [.000, .002]; DeepSeek stays cautious at 0.748 [.728, .767] (n=1925). Because the full OOS window is entirely thin, Compiled CE is identically zero by construction; thick-day economics are deferred to Sec. 5.4.

Blind arm. On the same 100 asset-days, the no-feed prompt *"Today is d. How should I trade BTC?"* yields abstention on 100% of days for all four models (CE = 0; zero directional actions). Without an observation layer, the models refuse. The costly failure mode is the middle case: the same model given a Raw fragment—half-baked context—starts trading (abstain as low as 0.04) and loses. Compiled matches Blind's refusal on thin days by the quality gate, while still exposing the structured world bundle that RQ2 and RQ3 show is usable when support improves.

Band-thick / open-slice days. Archive readiness alone is not a refuse policy. On full-OOS *band-thick* days (3/3 ready; 61% of OOS, n≈1224; ≡ WMI ≥ 0.05), production Compiled still abstains at 1.0 (WMI<0.2). Ungated on the same days keeps compiled content

Table 6: Full-OOS band-thick / open slice (3/3 ready ≡ WMI ≥ 0.05). Production Compiled abstains 1.0; Ungated CE point estimates are positive (bootstrap CIs include 0); Raw over-acts.

Model	Abs. U	CE U	Act-CE U	Abs. R	CE R
gpt-5.4-mini	0.58	+0.42	+0.32	0.00	−0.28
deepseek-v4-flash	0.42	+0.21	+0.45	0.75	+0.04
glm-5.2	0.58	+0.24	+0.11	0.09*	−0.87*
gemini-3.5-flash-lite	0.34	+0.31	+0.06	0.00	−0.25

*Raw GLM n=398; Ungated GLM n=932; others n=1224. Open-slice Ungated mean CE ≈+0.31 (block bootstrap B=5, 999 reps; one-sided p≈0.21 vs. CE ≤ 0).

without the hard boolean: mean abs. ≈0.48, CE positive for every vendor (mean ≈+0.29; Table 6). Block bootstrap (B=5, 999 reps) CIs include zero (mean CE ≈+0.31, one-sided p≈0.21); we report a content lower bound, not a trading edge, and *not* Compiled-open (the frozen prompt still forces abstain when the world is judged thin even if the boolean were false). Raw remains risky: GPT and Gemini abstain at 0.0 with CE −0.28 and −0.25.

Inside the Ungated arm. Ungated rationales cite disclosed world fields on 53–93% of days, so models *read* the observation layer's bundle; they differ in acting on it. Vendor-specific caution—not missing disclosure—drives the gap the hard quality gate closes.

Offline diagnostics. World-honoring offline followers abstain at 1.0 on full OOS; momentum-only mocks ignore thin-world flags (thin-abs. 0). Offline results are protocol checks; live vendors are the main claim.

When would the layer open? Compiled never trades under max WMI ≈ 0.093<0.2—a scarce-support stress test of the gate, not a permanently closed observation layer. A WMI gate of 0.05 would clear the boolean on n=1224 open days (ACWMI already ≥ 0.25), but would *not* clear the Compiled prompt's separate "world is thin" clause. Live evidence on that slice is Ungated content (Table 6) plus the no-LLM rule vs. momentum (ΔCE=0.315, p=0.034; Table 9). Denser vintages can raise the WMI floor without changing layer semantics.

5.3 RQ2: Seeing the market—grounding workflow

If the observation layer works, public LLMs should *see* what the world bundle discloses—not only obey a refuse bit. We run a live multi-step *grounding workflow* on 50 OOS asset-days per arm. Given Compiled or Raw inputs, the model must (i) list which of {exchange, macro, alternative} are ready, (ii) list which are missing, (iii) report sign(macro_tilt) and sign(alt_tilt), and (iv) judge sufficiency. Steps (i)–(iii) are scored against PIT ground truth. Table 7: with the Compiled bundle, mean ready-band F1 0.97, missing-band F1 0.97, and tilt-sign accuracy 0.97; with Raw, the same scores collapse to 0.04, 0.23, and 0.19. Sufficiency judgments alone are weak once completeness is scoped to real archive bands—models may call a 3/3-ready day "sufficient" while WMI still marks the world thin. The sharper claim is verifiability: only the compiled world bundle lets the model state what is ready, missing, and signed, in a form that can be checked. This is the analyst-facing face of the observation layer and the counterpart of Prop. 3.5's missingness term.

Table 7: Grounding workflow (50 days): ready / missing / tilt signs.

	Ready F1		Missing F1		Tilt-sign acc.	
Model	C	R	C	R	C	R
gpt-5.4-mini	0.94	0.00	0.94	0.31	0.94	0.20
deepseek-v4-flash	0.98	0.06	0.98	0.20	0.98	0.18
glm-5.2	0.98	0.09	1.00	0.11	0.98	0.20
gemini-3.5-flash-lite	0.96	0.03	0.96	0.30	0.96	0.18
Mean	0.97	0.04	0.97	0.23	0.97	0.19

Table 8: Secondary content probe: world-model rule vs. traditional baselines (no LLM).

Policy / contrast	Sharpe	CE
Buy-and-hold (always long)	−1.404	−1.163
Momentum always	0.096	−0.206
World-model content rule	**0.764**	**+0.130**
Mechanism − Momentum	—	0.334 (p=0.034)

5.4 RQ3: Secondary content probe

RQ1–RQ2 ask whether the observation layer is legible and safely gated. RQ3 asks whether the compiled tilts inside the world bundle are economically nonempty. A transparent rule shares return inputs with momentum and adds vintaged macro_tilt / alt_tilt—no LLM in the loop. Out-of-sample (199 days × 10 assets) this content rule beats the baselines (Table 8, Figure 4): Sharpe 0.764 / CE +0.130 versus momentum (0.096, −0.206) and buy-and-hold (−1.404, −1.163). The pre-specified contrast mechanism − momentum is significant under block bootstrap (ΔCE=0.334, p=0.034). ΔCE is positive for all ten assets (Figure 4, right). LOBO (Prop. 3.8) attributes the gain to band content (deleting macro/alternative collapses to momentum); gaps survive 10–25bps costs. A 2017–2026 audit with non-vintaged proxy tilts shows no edge (ΔCE ≈ 0, p=0.81): value appears where the vintage-compiled world exists. The result supports nonempty fields in the observation layer, not an LLM trading ranking.

Within-archive dense / open world. Among the three archive bands, 61% of OOS asset-days are band-thick (3/3 ready) and 39% are band-thin (exchange gap; Figure 2). On this panel, band-thick coincides with counterfactual open days (WMI ≥ 0.05). Table 9: on that open/thick slice the content rule acts and beats momentum (ΔCE=0.315; CE +0.061 vs. −0.254), stronger than on exchange-gap days (0.129)—complementary to RQ1's refuse-under-thin test.

5.5 Discussion and threats

What the results mean. Three facts carry the paper, in product order. (i) *The compiled world is legible:* grounding recovers ready/missing/tilt at ≈0.97 under Compiled vs. collapse under Raw (RQ2)—the observation layer makes the market visible to public LLMs. (ii) *Thin*

Table 9: Open/dense split of the content probe (OOS). Open = WMI $\geq 0.05 \equiv 3/3$ **bands ready.**

Slice	Share	Mech CE	ΔCE vs mom
All OOS	1.00	0.130	0.336
Open / thick (n=1224)	0.61	0.061	0.315
Thin (exchange gap)	0.39	1.048	0.129

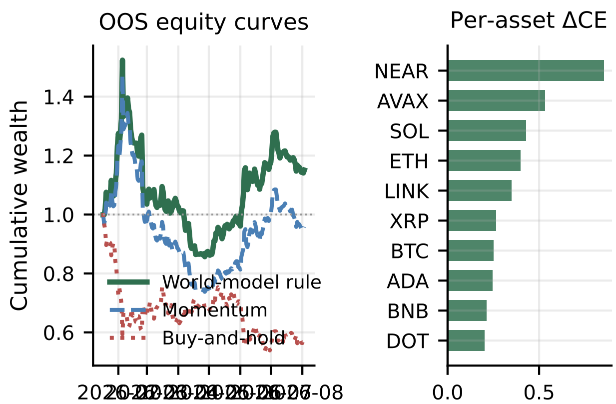

Figure 4: Secondary content probe (no LLM). Left: equity curves; right: ΔCE vs. momentum positive for every asset.

days must refuse: soft "consider abstaining" text does not; Compiled's hard quality gate does, across live vendors (RQ1). Disclosure without enforcement is unstable on thin mixes (Ungated ≈0.68; GLM 0.86 vs. Gemini 0.43) and often destroys CE versus cash when the model acts—half-baked context harms decisions. (iii) *Open-slice fields are a nonempty lower bound:* Ungated CE is positive for all four vendors on WMI ≥ 0.05 (Table 6; bootstrap CIs include zero), and RQ3's no-LLM rule is significant vs. momentum. Raw and Blind show unstructured or empty context cannot replace a typed world bundle. We claim a crypto observation layer—not LLM trading SOTA, not a generative simulator, not a full causal theory of crypto markets.

Relative to FinMem /FinAgent [23, 24] and ICAIF retrieval/debate/judge lines [1, 3, 18, 19], we contribute the observation compiler and quality gate those stacks assume, not a new memory/search/debate router. Relative to return-prediction LLM studies [14], we score seeing and safe refusal before PnL.

Design rationale. Three choices are deliberate. (i) Ship a world bundle, not a feature dump: completeness, honesty, and `should_ai_abstain` make quality machine-readable rather than a free-text apology. (ii) Previous-close PIT over wall-clock streaming: every live decision cites a recomputable `decision_asof`, avoiding same-day look-ahead. (iii) Four arms instead of one Sharpe number: Compiled / Ungated / Raw / Blind separates gated observation, soft judgment on the same bundle, thin integrations, and no feed.

Threats to validity. Compiled thin abstention is quality-gate obedience by design; Ungated removes the boolean on purpose. The archive is scarce (max WMI ≈ 0.093), so production Compiled never opens at 0.2. Open-slice Ungated is a content lower bound, not Compiled-open trading: the prompt's "world is thin" clause can still force abstain if the boolean were cleared, and CE bootstrap CIs include zero. Mixed-sample act-conditional Ungated CE is negative for three of four vendors; open-slice point estimates are positive—regime matters. Vendors are flash/mini at temperature 0; GLM Raw full-OOS is incomplete; denser bands are absent. Raw is common `mom5` wiring for the four-arm ladder, not dominance over tool-using agents.

Scope. We do not claim LLM trading SOTA or generative dynamics. Natural extensions include richer bands and thick-regime live trading once WMI leaves the scarce floor—decision/execution stacks can sit on top of this observation layer. Camera-ready will restore author identity and release the anonymized harness deferred during review.

6 Conclusion

Public LLMs need a crypto market they can safely observe before they decide. We contribute an observation layer: an anonymized state-compiler runtime that turns asynchronous multi-band evidence into a consumable PIT world bundle, and forces abstention when quality is insufficient so half-baked context cannot harm decisions. Live, the compiled bundle is legible (grounding ≈0.97 vs. Raw ≲0.23); the hard quality gate abstains 1.0 on every thin day while Ungated soft-judges at ≈0.68; Raw over-trades and loses; Blind refuses with no feed. Open-slice Ungated and a no-LLM content probe give a nonempty lower bound for the compiled fields, not a Compiled-open trading claim. The contribution is the observation layer—seeing plus quality-gated refusal—not a trading leaderboard.

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